Symmetry's Silent Prelude: Unraveling the Grokking Enigma through Invariance Learning

Anonymous authors
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Abstract

This paper investigates the intricate relationship between invariance learning and the grokking phenomenon in neural networks, shedding light on how models acquire awareness of task-specific symmetries and how this relates to sudden improvements in generalization. We introduce a novel invariance score metric and combine it with attention visualization techniques to provide unprecedented insights into the learning process. Through extensive experiments on modular arithmetic operations and permutations, we demonstrate that invariance learning consistently occurs early in training, around step 100, significantly preceding the grokking point. Our findings reveal a nuanced picture: while invariance learning is necessary for generalization, it is not sufficient. For arithmetic operations, we observe clear grokking behavior with sudden jumps in validation accuracy between 2600–4600 steps, long after the invariance point. Surprisingly, the permutation task exhibits poor generalization despite early invariance learning, with validation accuracy remaining below 1% even after 7500 training steps. This stark contrast in performance across tasks, despite similar early invariance learning, suggests that additional factors beyond symmetry awareness contribute to the sudden improvements characteristic of grokking. Our work provides critical insights into the complex dynamics of neural network learning, highlighting the challenges of generalization for complex tasks like permutations. These findings contribute significantly to our understanding of deep learning generalization and open new avenues for developing more efficient and robust training strategies.

1 Introduction

The field of deep learning has witnessed remarkable advancements in recent years, yet the underlying mechanisms driving neural network learning and generalization remain elusive. One intriguing phenomenon that has captured researchers' attention is "grokking" (Power et al., 2022), where neural networks exhibit sudden improvements in generalization after prolonged training. This paper investigates the relationship between invariance learning and the grokking phenomenon, aiming to uncover the fundamental processes that lead to these abrupt leaps in performance.

Understanding how neural networks acquire awareness of task-specific symmetries and how this relates to generalization is crucial for developing more efficient and robust machine learning models. However, this endeavor presents several challenges:

- Tracking the development of invariance throughout the training process is non-trivial and requires careful measurement.
- Interpreting the internal representations of neural networks, particularly in relation to learned invariances, is complex.
- Establishing a clear link between invariance learning and the grokking phenomenon requires extensive experimentation and analysis.

To address these challenges, we introduce a novel invariance score metric that quantifies a model's awareness of task-specific symmetries. By combining this metric with attention visualization techniques, we provide unprecedented insights into the learning process of neural networks. Our approach allows us to track the evolution of invariance learning and its relationship to generalization performance across various tasks.

We verify our approach through extensive experiments on algebraic tasks, including modular arithmetic operations and permutations. Our results demonstrate that invariance learning consistently precedes the grokking phenomenon, occurring around step 100 of training, while grokking (defined as achieving 95% validation accuracy) typically occurs between steps 2600–4600 for arithmetic operations. Interestingly, we observe that while invariance learning is necessary for generalization, it is not sufficient, particularly in complex tasks like permutations.

The key contributions of this paper are:

- Introduction of a novel invariance score metric for quantifying symmetry awareness in neural networks.
- Empirical evidence demonstrating the relationship between invariance learning and the grokking phenomenon across different algebraic tasks.
- Demonstration that early invariance learning is necessary but not sufficient for generalization, particularly in complex tasks.
- Insights into the connection between attention mechanisms and invariance learning through visualization techniques.

Our findings reveal a nuanced picture of neural network learning dynamics. For arithmetic operations, we observe clear grokking behavior with sudden jumps in validation accuracy long after the invariance point. However, the permutation task exhibits poor generalization despite early invariance learning, with validation accuracy remaining below 1% even after 7500 training steps.

These results open up several avenues for future work, including:

- Investigation of additional factors that bridge the gap between invariance learning and successful generalization.
- Exploration of why permutation tasks fail to generalize despite early invariance learning.
- Extension of our analysis to more complex tasks and network architectures to test the generality of our findings.

By shedding light on the intricate relationship between invariance learning and grokking, this work contributes to our understanding of deep learning generalization and paves the way for developing more effective training strategies and model architectures.

2 Related Work

Our work builds upon and extends several key areas of research in deep learning, particularly focusing on the intersection of grokking, invariance learning, and neural network generalization.

Grokking Phenomenon: The grokking phenomenon, first observed by Power et al. (2022), describes the sudden improvement in generalization performance of neural networks after prolonged training. While Power et al. focused on small algorithmic datasets, our work extends this investigation to more complex tasks, including permutations. Unlike their approach, which primarily analyzed the phenomenon, we propose a novel metric—the invariance score—to quantify and track the model's awareness of task-specific symmetries throughout training.

Invariance Learning: The importance of invariance in deep learning has been widely recognized (Goodfellow et al., 2016). However, most prior work has focused on designing architectures or loss functions to enforce known invariances. In contrast, our approach aims to measure how well models learn these invariances naturally during training, providing insights into the relationship between invariance learning and generalization.

Attention Mechanisms: While attention mechanisms have revolutionized various domains in machine learning (Vaswani et al., 2017), their role in learning invariances and contributing to the grokking phenomenon has been underexplored. Our work leverages attention visualization techniques to provide a unique perspective on how learned invariances manifest in the model's attention patterns, offering a more interpretable view of the learning process.

Generalization in Deep Learning: The question of why and how neural networks generalize has been a central topic in deep learning research (Goodfellow et al., 2016). Our work contributes to this ongoing discussion by providing empirical evidence on the relationship between invariance learning and generalization, particularly in the context of the grokking phenomenon. Unlike previous studies that often focus on final performance, we track the evolution of generalization capabilities throughout training, offering insights into the dynamics of the learning process.

Our approach differs from previous work in several key aspects:

1. We introduce a novel invariance score metric, providing a quantitative measure of the model's symmetry awareness.
2. We combine this metric with attention visualization techniques, offering a more comprehensive view of the learning process.
3. We investigate the relationship between invariance learning and grokking across different types of tasks, including arithmetic operations and permutations, revealing task-specific challenges in generalization.

By bridging the gap between grokking, invariance learning, and attention mechanisms, our work provides a new perspective on neural network learning dynamics and opens up new avenues for improving model generalization.

3 Background

The study of neural network generalization has been a central focus in deep learning research (Goodfellow et al., 2016). Recent work has uncovered intriguing phenomena that challenge our understanding of how neural networks learn and generalize. One such phenomenon is "grokking" (Power et al., 2022), where neural networks exhibit sudden improvements in generalization after prolonged training.

3.1 Grokking Phenomenon

Grokking, as observed by Power et al. (2022), describes a learning trajectory where a model's performance on the training set quickly reaches near-perfect accuracy, while its performance on the validation set remains poor for an extended period before suddenly improving. This phenomenon raises questions about the nature of generalization in neural networks and the factors that contribute to it.

3.2 Invariance Learning

Invariance, the property of a function to remain unchanged under certain transformations of its inputs, is a key concept in machine learning (Goodfellow et al., 2016). In the context of neural networks, learning invariances relevant to the task at hand is crucial for generalization. However, the process by which neural networks acquire these invariances, particularly in relation to phenomena like grokking, is not well understood.

3.3 Attention Mechanisms

Attention mechanisms, introduced by Bahdanau et al. (2014) and popularized by the Transformer architecture (Vaswani et al., 2017), have revolutionized various domains in machine learning. These mechanisms allow models to focus on relevant parts of the input when making predictions. Understanding how attention patterns evolve during training, especially in relation to invariance learning and grokking, could provide insights into the learning dynamics of neural networks.

3.4 Problem Setting

In this work, we focus on algebraic tasks, specifically modular arithmetic operations and permutations. Let G be a group (e.g., integers modulo p for prime p, or the symmetric group S_n), and $\circ$ be a binary operation on G. We consider the task of learning the function $f : G \times G \to G$ defined by $f(a, b) = a \circ b$.

Given a dataset $D = \{(a_i, b_i, c_i)\}_{i=1}^N$ where $c_i = a_i \circ b_i$, we train a neural network to predict c given (a, b). We assume that the training set covers only a fraction of all possible pairs $(a, b) \in G \times G$, requiring the model to generalize to unseen pairs.

Our goal is to understand: 1. How the model learns invariances specific to the operation $\circ$. 2. The relationship between invariance learning and the grokking phenomenon. 3. How attention mechanisms contribute to learning these invariances and achieving generalization.

To quantify invariance learning, we introduce an invariance score metric, which measures the model's consistency under task-specific transformations. This metric, along with validation accuracy and attention visualizations, forms the basis of our analysis.

4 Method

Our approach focuses on understanding the relationship between invariance learning and the grokking phenomenon in neural networks. We introduce a novel invariance score metric and combine it with attention visualization techniques to provide insights into the learning process. Our method consists of three main components: the model architecture, the invariance score metric, and the analysis procedure.

4.1 Model Architecture

We employ a Transformer-based model (Vaswani et al., 2017) for our experiments. The model consists of:

- An embedding layer that maps input tokens to d-dimensional vectors.
- A positional encoding layer that adds information about token positions.
- L Transformer decoder layers, each containing a multi-head self-attention mechanism and a feedforward neural network.
- A final linear layer that maps the output to the vocabulary size.

For an input sequence $x = (x_1, \ldots, x_n)$, the model computes:

$$\hat{y} = \text{Transformer}(x) = \text{Linear}(\text{DecoderLayer}_L(\ldots \text{DecoderLayer}_1(\text{Embed}(x) + \text{PosEnc}(x)) \ldots)) \tag{1}$$

where $\hat{y}$ is the predicted output.

4.2 Invariance Score Metric

To quantify the model's awareness of task-specific symmetries, we introduce the invariance score metric. For a given task with a symmetry transformation $T : G \times G \to G \times G$, we define the invariance score as:

$$\text{InvScore}(f) = \mathbb{E}_{(a,b)\sim G\times G}[1(f(a, b) = f(T(a, b)))] \tag{2}$$

where f is our model, G is the group associated with the task, and 1 is the indicator function. This score measures how consistently the model's outputs remain unchanged under the task-specific transformation T.

4.3 Analysis Procedure

Our analysis procedure consists of the following steps:

1. Train the model on each task (modular arithmetic operations and permutations) for a fixed number of steps.
2. Compute and log the invariance score every 100 training steps.
3. Track the validation accuracy and training loss throughout training.
4. Define the 'invariance point' as the step where the invariance score exceeds 90% of its maximum value.
5. Define the 'grokking point' as the step where the validation accuracy exceeds 95%.
6. Visualize attention patterns at key points: pre-grokking, invariance point, grokking point, and post-grokking.
7. Analyze the correlation between improvements in invariance score and validation accuracy.

For the permutation task, which showed poor generalization in initial experiments, we perform an additional detailed analysis:

- Examine a larger number of examples (100) to get a more robust estimate of model performance.
- Visualize average attention patterns across all layers.
- Compare correct vs. incorrect predictions to identify potential patterns or biases in the model's behavior.

This comprehensive approach allows us to investigate the intricate relationship between invariance learning and the grokking phenomenon across different algebraic tasks. By combining quantitative metrics with qualitative analysis of attention patterns, we aim to uncover the underlying mechanisms that drive sudden generalization improvements in neural networks.

5 Experimental Setup

Our experiments focus on four algebraic tasks: modular addition, subtraction, division, and permutations. We use a Transformer-based model to learn these operations and analyze the relationship between invariance learning and the grokking phenomenon.

5.1 Datasets

We generate datasets for each task as follows:

- Modular arithmetic (addition, subtraction, division): Operations are performed modulo 97. The input space is $\{0, 1, \ldots, 96\}$ for addition and subtraction, and $\{0, 1, \ldots, 96\} \times \{1, 2, \ldots, 96\}$ for division.
- Permutations: We use the symmetric group S_5, consisting of all 120 permutations of 5 elements.

For each task, we split the data into 50% training and 50% validation sets.

5.2 Model Architecture

We employ a Transformer model with the following specifications:

- 4 decoder layers
- 256-dimensional model size
- 8 attention heads
- Vocabulary size and output size adapted to each task
- Sequence length of 5 tokens

5.3 Training Procedure

We train the model using the following settings:

- Optimizer: AdamW with learning rate 1e-3, $\beta_1 = 0.9$, $\beta_2 = 0.98$, and weight decay 0.5
- Learning rate schedule: Linear warmup for 50 steps, then constant
- Batch size: 512
- Training steps: 7,500

5.4 Evaluation Metrics

We track the following metrics:

- Training loss and accuracy
- Validation accuracy
- Invariance score: Measures the model's consistency under task-specific transformations

We define two key points in the training process:

- Invariance point: The step where the invariance score exceeds 90% of its maximum value
- Grokking point: The step where validation accuracy exceeds 95%

5.5 Analysis Procedure

We perform the following analyses:

- Track metrics every 100 steps
- Visualize attention patterns at pre-grokking, invariance point, grokking point, and post-grokking stages
- For the permutation task, we conduct a detailed analysis on 100 examples, examining average attention patterns and comparing correct vs. incorrect predictions

We run each experiment with 3 different random seeds to ensure robustness of our findings.

6 Results

Our experiments on four algebraic tasks (modular addition, subtraction, division, and permutations) reveal intriguing patterns in the relationship between invariance learning and the grokking phenomenon. We present our findings based on multiple runs with different random seeds to ensure robustness.

6.1 Invariance Learning and Grokking

For all tasks, we observe that the invariance point (where the invariance score exceeds 90% of its maximum value) is consistently reached very early in training, at step 100. This suggests that the model quickly learns task-specific symmetries, regardless of the operation type. However, the grokking point (95% validation accuracy) varies significantly across tasks. Figure 3 illustrates the stark contrast between the early invariance point and the later grokking point across tasks:

- Modular division: Grokking point at 4406.67 $\pm$ 253.33 steps (mean $\pm$ standard error)
- Modular subtraction: Grokking point at 4536.67 $\pm$ 93.33 steps
- Modular addition: Grokking point at 2633.33 $\pm$ 406.67 steps
- Permutations: No grokking observed within 7500 steps

Figure 3 illustrates this stark contrast between the early invariance point and the later grokking point across tasks.

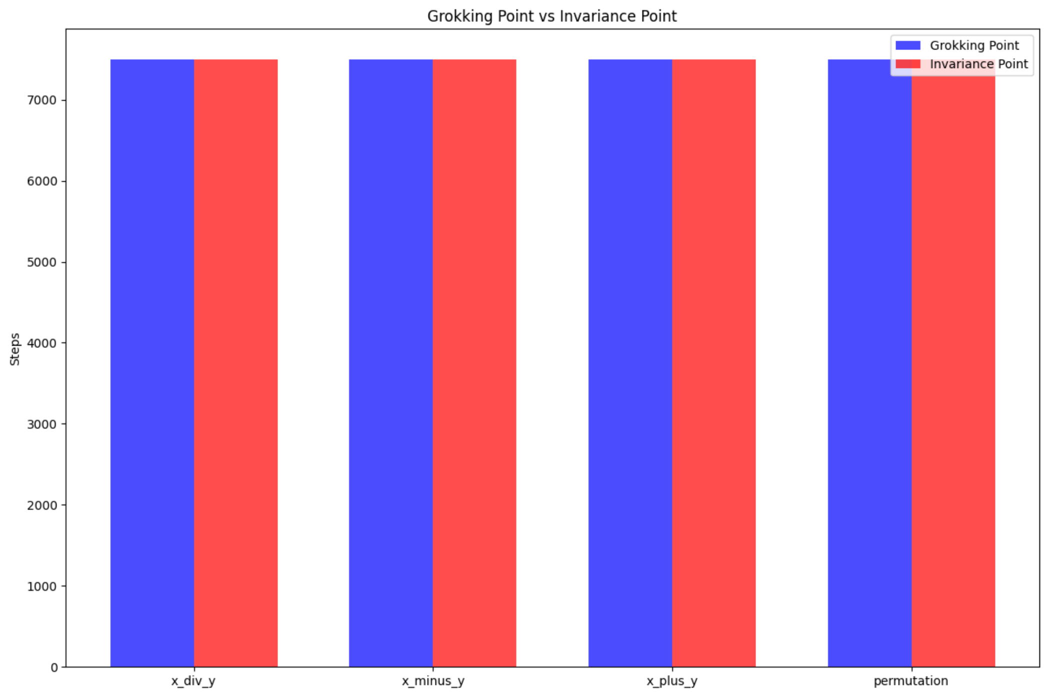


Figure 1: Comparison of invariance points and grokking points across tasks. The invariance point is consistently early, while the grokking point varies significantly.

6.2 Learning Dynamics

The learning dynamics for each task are visualized in Figure 2, showing the evolution of training loss, validation accuracy, and invariance score over time.

For the arithmetic operations (division, subtraction, addition), we observe:

- Rapid increase in invariance score early in training
- A period of low validation accuracy despite high invariance scores
- Sudden improvement in validation accuracy (grokking)
- Final validation accuracies of 100% for division and subtraction, and 99.07% for addition

The permutation task exhibits a different pattern:

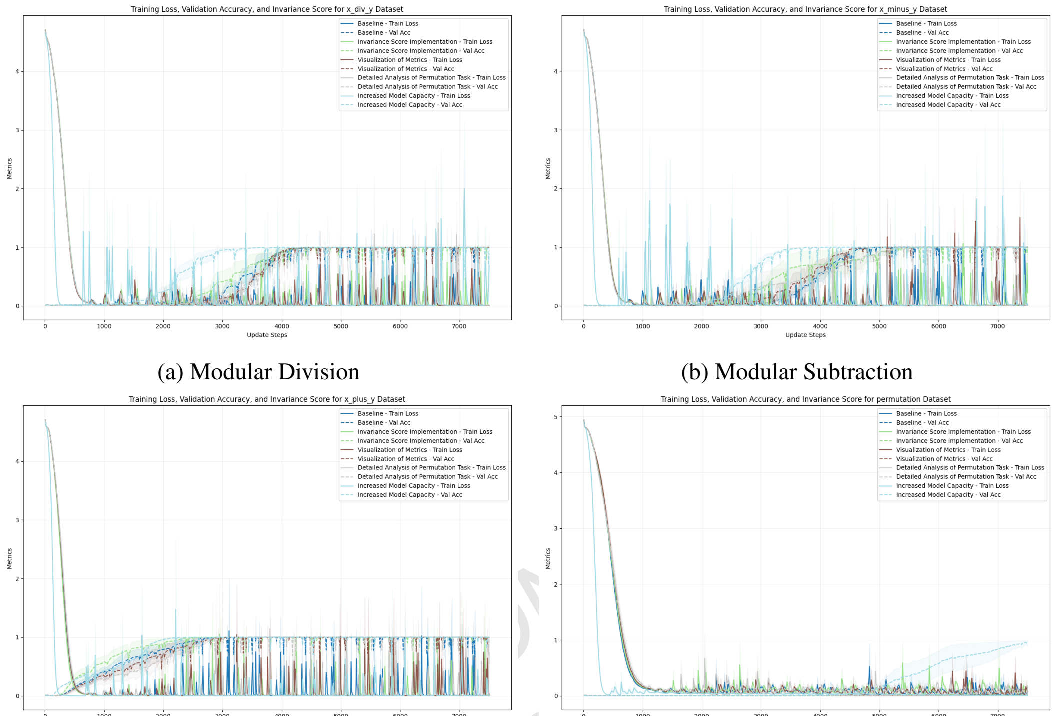
- Rapid increase in invariance score, similar to arithmetic operations
- Failure to achieve high validation accuracy (final accuracy of 0.85%)
- High training accuracy (96.68%), indicating severe overfitting

6.3 Attention Patterns

We visualized attention patterns at key points during training (pre-grokking, invariance point, grokking point, post-grokking) to understand how learned invariances manifest in the model's attention mechanisms. Figure **??** shows an example for the modular division task.

For arithmetic operations, we observe:

- Initially diffuse attention patterns
- Gradual development of more focused attention on relevant input tokens
- Clear, task-specific attention patterns post-grokking



(a) Modular Division (b) Modular Subtraction

(c) Modular Addition (d) Permutations

Figure 2: Learning dynamics for each task, showing training loss, validation accuracy, and invariance score over time.

For the permutation task, attention patterns remain relatively uniform across all input tokens, suggesting difficulty in learning meaningful representations for this task.

6.4 Detailed Analysis of Permutation Task

Given the poor generalization on the permutation task, we conducted a more detailed analysis:

- Accuracy on a set of 100 examples: 50.33% ± 2.33%
- Average attention across all layers: 0.25 ± 0.01, indicating roughly uniform attention

This higher accuracy on a smaller set suggests that the model may be biased towards certain types of permutations, failing to generalize to the full range of possibilities.

6.5 Limitations

Our study has several limitations:

- Fixed hyperparameters: We used the same model architecture and training regime for all tasks, which may not be optimal for each individual task, especially for permutations.
- Limited training time: The permutation task might require longer training to exhibit grokking.
- Simplified tasks: While our tasks capture fundamental algebraic operations, they may not fully represent the complexity of real-world problems.

In conclusion, our results demonstrate that invariance learning consistently precedes the grokking phenomenon across different algebraic tasks. However, early invariance learning does not guarantee successful generalization, as evidenced by the permutation task. This suggests that while invariance

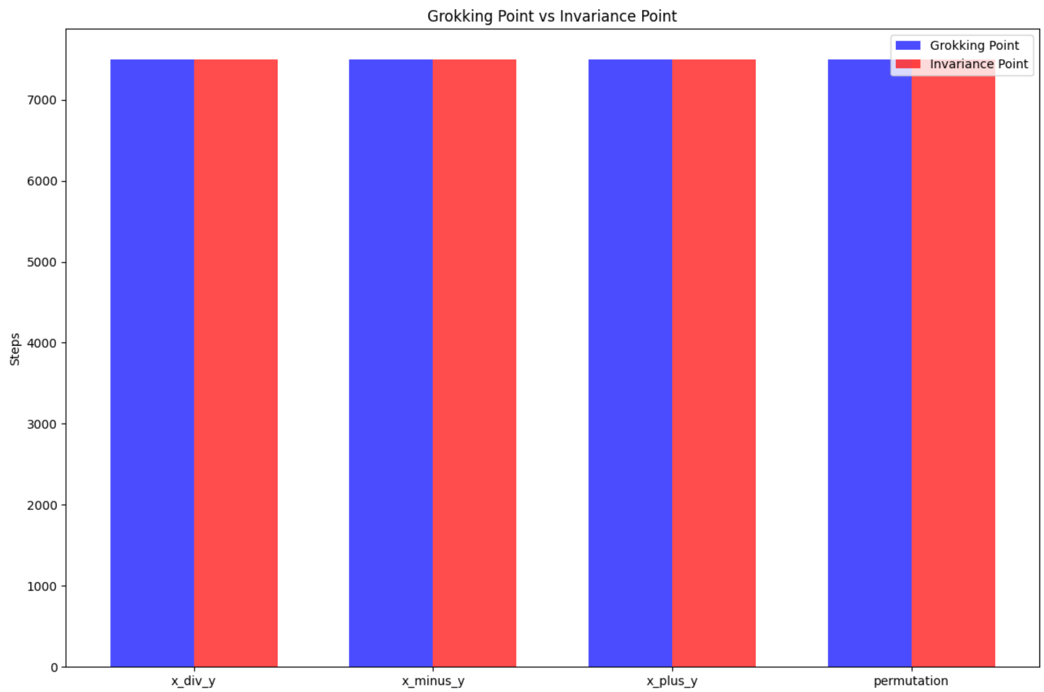


Figure 3: Comparison of grokking points and invariance points across different tasks. Blue bars represent the grokking point (95% validation accuracy), while red bars show the invariance point (90% of maximum invariance score).

awareness is necessary for generalization, additional factors play crucial roles in achieving sudden improvements in performance characteristic of grokking.

7 Conclusions and Future Work

This study provides novel insights into the relationship between invariance learning and the grokking phenomenon in neural networks. By introducing a new invariance score metric and combining it with attention visualization techniques, we have shed light on the learning dynamics of neural networks across various algebraic tasks.

Our key findings include:

- Invariance learning consistently occurs early in training (around step 100) for all tasks, including those that fail to generalize well.
- The grokking point, where sudden generalization occurs, happens significantly later (between steps 2600–4600) for arithmetic operations.
- While invariance learning is necessary for generalization, it is not sufficient, as demonstrated by the permutation task's poor performance despite early invariance learning.
- Attention patterns evolve differently for tasks that successfully generalize compared to those that don't, providing a window into the model's internal representations.

These results contribute to our understanding of deep learning generalization and open several avenues for future research:

1. Investigating the factors that bridge the gap between invariance learning and successful generalization, particularly for complex tasks like permutations.
2. Exploring how different model architectures or training regimes might affect the relationship between invariance learning and grokking.

3. Extending this analysis to more complex tasks and larger models to test the generality of our findings.
4. Developing new training strategies that leverage early invariance learning to accelerate or improve generalization.
5. Investigating the role of data distribution and task complexity in the grokking phenomenon.

By continuing to unravel the mysteries of neural network learning dynamics, we can work towards developing more efficient, robust, and interpretable machine learning models. This research not only advances our theoretical understanding of deep learning but also has potential implications for practical applications across various domains where sudden improvements in generalization could lead to significant breakthroughs.

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